

MIT 6.087 — Lecture 3: Control Flow & Conditionals

Practical Programming in C • Conditional Structures, Branching & Selection Logic

A simple statement in C ends with a semicolon (;).

1. Blocks and Compound Statements

Multiple statements enclosed within curly braces { ... } form a compound statement or block.

A block can substitute for any simple statement and is treated by the compiler as a single execution unit.

Unlike higher-level languages with dedicated boolean keywords, in C (C89/C90), any non-zero numeric expression is considered TRUE, and zero (0) is considered FALSE.

In C99, the bool type is provided via <stdbool.h>.

3. The if-else and Nested if Statements

```
if (x % 2 == 0) {  
    y += x / 2;  
} else if (x % 4 == 1) {  
    y += 2 * ((x + 3) / 4);  
} else {  
    y += (x + 1) / 2;  
}
```

The switch Statement & Iteration Loops

1. The switch-case Multi-Branch Construct

An alternative conditional statement taking an integer or character variable as input. Cases fall through unless explicitly terminated using the break keyword.

```
switch (ch) {  
    case 'Y':  
    case 'y':  
        printf("Confirmed\n");  
        break;  
    case 'N':  
        printf("Declined\n");  
        break;  
    default:  
        printf("Invalid input\n");  
        break;  
}
```

2. Loop Iterations: while, for, and do-while

- while (condition) { ... } : Entry-controlled loop; condition evaluated first.
- for (init; cond; incr) { ... } : Ideal for counting iterations.
- do { ... } while (condition); : Exit-controlled loop; body executes at least once!
- break : Exits the innermost loop immediately.
- continue : Skips remaining body statements and jumps to the next loop iteration.